

Product Requirements Document

AxionHR — Workplace Wellbeing Ecosystem

Hackathon: AxionHR Wellness & Mood Quality Hackathon

Submission Stage: Stage 2 — Blueprint Design

Team: HTTP hack

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1. Overview and Objective

AxionHR is one connected ecosystem for workplace wellbeing, built for the AxionHR Wellness & Mood Quality Hackathon. Instead of six disconnected point solutions, all six challenge pillars share one employee data core, one design system, and one codebase — so a signal from one pillar can inform another, culminating in a single unified employee profile.

The Six Pillars

#	Pillar	Brief Category	Status
1	Burnout Risk Analytics	Predictive Analytics	Designed
2	Quick Nudge Tool	Physical Health	Designed
3	Track the Mood	Mental Well-being	Designed
4	Boundary Boundary ("Right to Disconnect")	Work-Life Boundaries	Designed
5	Give me a Coffee ("Random Buddy Check-ins")	Social Health	Designed
6	Adapt to the Workspace ("Calendar-Driven Focus Mode")	Cognitive Inclusivity	Designed

Objective: AxionHR catches burnout and disengagement before they turn into resignations, solving the four failures of traditional wellness tools (slow surveys, invisible burnout, eroding boundaries, disconnected point tools) by unifying six wellbeing features into one system where a signal in one pillar (like rising burnout risk) can inform another (like triggering a calmer workspace mode), giving managers and HR a real-time picture of team wellbeing instead of a quarterly snapshot.

2. Background and Context

Traditional HR wellness tooling relies on infrequent engagement surveys, isolated point tools, and top-down recognition programs that don't talk to each other. The hackathon brief poses six discrete challenge categories, Predictive Analytics, Physical Health, Mental Well-being, Work-Life Boundaries, Social Health, and Cognitive Inclusivity. Rather than build six separate apps, the team chose to unify them under one product architecture: one repository, one Supabase project, and a shared employee/activity core that every pillar reads from and writes into.

As of this document, all six pillars are at the Designed stage, the team has spaced out the logic, data model, and screens for each pillar, but none have been built yet.

3. Problem Statement

Traditional HR wellness tooling fails in four specific ways this product addresses:

- Annual/quarterly engagement surveys are lagging indicators, sentiment is captured months after it mattered, and participation suffers from survey fatigue.
- Burnout goes unnoticed until an employee resigns or takes extended leave, there is no early, non-invasive signal today.
- Remote/hybrid work has eroded the boundary between office hours and personal time, producing toxic "always-on" cultures.

- Point solutions (a survey tool, a wellness app, a calendar plugin) don't talk to each other, so no one gets a holistic picture of team health.

4. Users and Personas

Personas are derived directly from each pillar's stated privacy boundaries for example, the Mood pillar guarantees that "no code path anywhere joins mood value back to an individual's name," which is why the Manager and HR personas below only ever see aggregate views, never individual-level data.

Persona	Description	Primary Interactions
Employee (Individual Contributor)	The primary daily user; every pillar is built around their workday.	Receives nudges; logs one-tap daily mood; sends/receives boundary-aware messages; is paired daily for buddy check-ins; experiences adaptive Focus/Calm UI states.
People Manager / Team Lead	Oversees a team's wellbeing at an aggregate level only.	Views the Team Overview burnout table and band summary; views team-level mood trend (never individual-level).
HR Administrator	Handles workforce-wide wellbeing follow-up.	Reviews the unlinked "flagged for HR" queue from buddy check-ins; relies on aggregate, privacy-preserving signals rather than named individual data.

5. Goals and Success Metrics

Goal	Success Metric
Ship a centralized wellbeing platform, not six disconnected tools.	All 6 pillars share the same employee_id-keyed core (employees, daily_activity) and one Supabase project.
Every scoring/decision engine is explainable, not a black box.	Burnout, Boundary, and Focus logic remain fully rule-based and expressible in one sentence each.
Every score or decision is proven, not eyeballed.	Automated tests pass against the Burnout archetypes/edge cases and the Boundary timing scenarios before merge.
Privacy-by-design holds under test, not just in the UI.	No individual mood value, buddy check-in, or focus-mode state is ever exposed outside its owner (or, for flags, HR) at the query level.
The "ecosystem" story is demonstrable.	Centralized Employee Profile ships once at least 2-3 pillars are built, joining their data into one view.
Stage 2 (Blueprint Design) deliverable is complete on time.	This PRD plus hi-fi prototypes for all six pillars, using the shared design tokens, submitted by the deadline.

6. Scope and MVP

All six pillars are currently at the Designed stage, none have been built yet. Scope below is organized by build priority.

MVP, most fully specified, build first

- Shared core schema (employees, daily_activity) over a synthetic dataset: 20 employees across 4 teams, 30 days of activity history.
- Burnout Risk Analytics — full scoring engine spec, Team Overview + Employee Detail screens, 3 API endpoints, validation plan against 4 archetypes plus 2 edge cases.

- Quick Nudge Tool — full trigger logic spec, live demo screen, validation plan.

Near-term scope — build next

- Track the Mood — 2 screens, frictionless one-tap check-in.
- Boundary Boundary — 2 screens, simulated compose/send tool.

Later scope — build after the above

- Give me a Coffee (Random Buddy Check-ins) — 3 screens.
- Adapt to the Workspace (Focus Mode) — 2-3 screens, including the first cross-pillar data read (consumes Burnout's risk_scores.band).

Explicitly out of scope

- Centralized Employee Profile — deferred until 2-3 pillars are built.
- Real Slack/Email/Calendar API integrations for Boundary and Focus.
- Individual mood history view — only aggregate trends are in scope.
- Any point-scoring or leaderboard mechanic for peer recognition.

7. Functional Requirements

7.1 Burnout Risk Analytics — Designed, not yet built

Burnout is invisible until someone resigns; surveys are lagging indicators. This pillar scores burnout risk from behavioral signals already in the shared core.

- **FR-1.1** Compute four sub-scores per employee over a 30-day rolling window: streak, meeting-load, off-hours-messaging, and days-since-PTO.

```
streakScore = min(100, streak_days x 8)
meetingScore = avg(meeting_hours / available_hours) x 100, capped 100
offHoursScore = min(100, (avg_weekly_off_hours_messages / 15) x 100)
ptoScore = min(100, days_since_pto / 1.2), +30 flat penalty if messaged while on_pto = true
```

- **FR-1.2** Combine sub-scores into a weighted composite score (30% streak / 25% meeting / 25% off-hours / 20% PTO) and map it to one of four bands: Low (0-24), Medium (25-49), High (50-74), Critical (75-100).
- **FR-1.3** Expose GET /api/burnout/employees, GET /api/burnout/employees/[id], and POST /api/burnout/recompute.
- **FR-1.4** Validate the engine against four archetypes (Healthy Hannah/Low, Warning Will/Medium, Risky Rita/High, Burnout Bob/Critical) plus 2 edge cases (zero data; every factor maxed).
- Screens: Team Overview (sortable table, band-count summary strip); Employee Detail (4 sub-score bars, 14-day trend line, plain-language explanation).

Status: Designed, not yet built

7.2 Quick Nudge Tool — Designed, not yet built

Continuous desk work causes eye strain, poor posture, and sedentary habits. This pillar nudges employees toward small physical resets without becoming annoying.

- **FR-2.1** Trigger a nudge when the active session exceeds 50 minutes and no meeting starts in the next 10 minutes.
- **FR-2.2** Rotate nudge type (stretch -> hydrate -> eye-rest -> posture) without repeating the same type back-to-back.
- **FR-2.3** Suppress nudges during quiet hours (default 6pm-8am) and once the daily cap (default 6/day) is reached.
- **FR-2.4** Support snoozing a nudge (snoozed_until) so it does not re-fire early.

- Screens: Live demo — accelerated work-session simulator, toast-style non-blocking nudge cards (Done / Snooze 10 min), today's nudge log.

Status: Designed, not yet built

7.3 Track the Mood — Designed, not yet built

Annual/quarterly surveys have low participation and miss the daily/weekly pulse.

- **FR-3.1** One-tap check-in across five emoji values (1-Awful to 5-Great), completed in under 5 seconds, with no note field and no extra steps.
- **FR-3.2** Enforce one check-in per employee per day via upsert, not duplicate rows.
- **FR-3.3** Enforce at the query level — not just the UI — that individual check-ins are never returned by employee name; only team/day-level averages are ever queryable.
- Screens: Daily Check-in (single screen, 5 emoji taps); Team Trends (14-day average per team, with a visible privacy statement on the page).

Status: Designed, not yet built

7.4 Boundary Boundary ("Right to Disconnect") — Designed, not yet built

Hybrid/remote work blurs office hours vs. personal time, feeding an always-on culture.

- **FR-4.1** Provide a simulated compose-and-send tool (sender/recipient chosen from the synthetic employee list) rather than a live email/Slack integration.
- **FR-4.2** Evaluate every send at send time: in-hours + not on PTO -> Delivered; outside hours + not on PTO -> Delayed to the recipient's next working-hours start; recipient on PTO -> Warned (names the return date), with an explicit "send anyway" override. The system warns, it never hard-blocks.
- **FR-4.3** Log every send with its resulting action (delivered / delayed / warned_sent_anyway / blocked_until_return) and, where applicable, the scheduled delivery time.
- **FR-4.4** Validate against four scenarios: in-hours, after-hours, on-PTO, and exactly at the working-hours boundary.
- Screens: Compose/Send Simulator (pick sender + recipient, write a message, see the resulting action explained in plain language); Recent Activity (feed of past intercepted messages and outcomes).

Status: Designed, not yet built

7.5 Give me a Coffee ("Random Buddy Check-ins") — Designed, not yet built

Isolation in remote work; top-down recognition feels forced; most peer-recognition tools reinforce existing cliques instead of solving isolation.

- **FR-5.1** Pair the whole team 1:1 daily via round-robin rotation (circle method) so no pairing repeats until every teammate has connected once (one full cycle, e.g. "2026-Q3").
- **FR-5.2** Track progress as cycle completion (e.g., "8/24 teammates connected this quarter"), never as a per-person leaderboard; a pairing only counts once both people submit a check-in (status = completed).
- **FR-5.3** Offer a check-in prompt of one predefined tag (Great teammate / Made my day / Went above and beyond / Really listened) plus an optional free-text note.
- **FR-5.4** Keep check-ins private between the pair by default, with no public leaderboard, ever.
- **FR-5.5** Provide an optional, quiet "I think HR should know how my buddy's doing" flag, surfaced only in the unlinked HR view — how this pillar covers anonymous concern-flagging without a bolted-on separate button.
- **FR-5.6** Tie rewards to participation and team-wide milestones only, never to individual popularity.
- Screens: Today's Buddy (pairing, tag + note prompt, submit); My Progress (cycle-completion stat); HR view, unlinked from main nav (flagged entries for review).

Status: Designed, not yet built

7.6 Adapt to the Workspace ("Calendar-Driven Focus Mode") — Designed, not yet built

Standard HR/workflow tools are built for neurotypical workers; employees with ADHD, autism, or anxiety struggle with notification overload and rigid scheduling.

- **FR-6.1** Support three UI states — Standard (full dashboard/notifications/task list); Focus/Simplified (batched notification count only, top-3 priority tasks only, calmer visuals); Calm/Recovery (notifications release gradually, layout eases back toward Standard).
- **FR-6.2** Trigger Focus/Simplified from calendar signals: a marked Deep Work block, 2+ back-to-back meetings with no gap, or an event tagged high-stress.
- **FR-6.3** Use the employee's current Burnout band (`risk_scores.band`) as a baseline-sensitivity input — a Critical-band employee needs less calendar "evidence" to trigger Focus/Simplified than a Low-band employee. This is the first pillar to read another pillar's data, and the architecture's proof point that the shared core works.
- **FR-6.4** Trigger Calm/Recovery in a gap between meetings, or right after a stressful stretch ends.
- **FR-6.5** Always provide a manual start/stop override regardless of what any signal says, and always surface why the current state activated.
- **FR-6.6** Keep focus-mode state and sensitivity settings fully private to the individual — not visible to managers or anyone else.
- Screens: Focus Dashboard (state indicator + manual control); Standard vs. Focus/Simplified comparison view; optional day timeline of `calendar_events` blocks and state transitions.

Status: Designed, not yet built

8. Non-Functional Requirements

- Privacy-by-design throughout, not only in Mood: aggregate-only mood data, private buddy check-ins with no leaderboard, individual-only focus settings, and boundary warnings that inform rather than surveil.
- Transparency over black-box ML: every scoring/decision system (Burnout, Boundary, Focus) is rule-based and explainable in one sentence — no opaque model in the loop.
- Validated, not just demoed: anywhere a score or decision exists, it is proven against known-outcome test cases before it ships, not just eyeballed.
- No real third-party integrations: everything runs on synthetic data (20 employees, 4 teams, 30 days of history) to keep hackathon build risk low and demo reliability high — a deliberate scope decision, not a limitation to hide.

9. Data Requirements

All six pillars key off two shared core tables. Each pillar owns its own tables in its own numbered migration file, so parallel work never collides.

Shared Core Schema

```
employees: id, name, team, role, email, created_at
daily_activity: id, employee_id, date, meeting_hours, available_hours,
                off_hours_messages, worked_today, on_pto, created_at
                (unique per employee + date)
```

Migration Ownership

Migration	Owns
0001_foundation.sql	employees, daily_activity (shared core)
0002_burnout.sql	risk_scores
0003_nudge.sql	nudge_events, nudge_preferences
0004_boundary.sql	boundary_events
0005_kudos.sql	buddy_pairings, buddy_checkins
0006_mood.sql	mood_checkins
0007_focus.sql	focus_sessions, calendar_events

Per-Pillar Data Models

```
risk_scores(employee_id, computed_date, streak_score, meeting_score,
            off_hours_score, pto_score, composite_score, band)
nudge_events(employee_id, nudge_type, triggered_at, acknowledged, snoozed_until)
nudge_preferences(employee_id, quiet_hours_start, quiet_hours_end, max_nudges_per_day)
mood_checkins(employee_id, date, mood_value 1-5) -- unique per employee+date, no note column
boundary_events(sender_id, recipient_id, channel, message_preview, sent_at,
               action: delivered|delayed|warned_sent_anyway|blocked_until_return,
               scheduled_delivery)
buddy_pairings(id, employee_a_id, employee_b_id, cycle_id, round_number, date,
               status: pending|partial|completed|expired)
buddy_checkins(id, pairing_id, employee_id, tag, note, flagged_for_hr, flagged_note, completed_at)
focus_sessions(employee_id, date, start_time, end_time, notifications_batched, deep_work)
```



```
calendar_events(employee_id, date, start_time, end_time,
                 event_type: meeting|deep_work|high_stress, label)
```

cycle_id is a simple label (e.g. "2026-Q3") generated when a rotation starts — no separate cycle-tracking table. The round-robin schedule itself is computed by a script at pairing-generation time, not modeled as its own entity.

10. Integrations and Dependencies

- No real third-party integrations: Slack, Calendar, and Email are all simulated against the synthetic employee list — explicit, deliberate scope, not a gap to hide.
- Internal cross-pillar dependency: Adapt to the Workspace reads Burnout Risk Analytics' risk_scores.band as a baseline-sensitivity input — Focus Mode cannot be finished, tested, or demoed correctly until Burnout's band output is stable.
- Shared infrastructure dependency: one repository (axionhr), one Supabase project — every pillar depends on the shared core migration (0001_foundation.sql) landing first.
- Team process dependency: main is protected; each pillar builds on its own feature branch (feature/burnout, feature/nudge, etc.) and merges only after that pillar's own build/tests pass; branches must pull main before starting new work to avoid migration-file collisions.
- Library dependencies: Next.js 14 (App Router), TypeScript, Tailwind CSS, and Supabase (Postgres) form the shared stack; recharts renders all trend/score charts and lucide-react supplies iconography, kept consistent across pillars.

11. UX/Design and Brand

Design tokens (Tailwind theme extension) are fixed and are meant to carry unchanged from the hi-fi prototype into the actual build, so the two never diverge.

Token	Hex	Meaning
risk-low	#22C55E	Burnout band: Low
risk-medium	#EAB308	Burnout band: Medium
risk-high	#F97316	Burnout band: High
risk-critical	#EF4444	Burnout band: Critical
brand	#0EA5E9	Primary brand color
brand-dark	#0369A1	Primary brand color, dark variant

Typeface: Inter, used across all six pillars for visual consistency.

Design principle (derived from the pillars' explicit UX decisions): across every pillar, the interface leans supportive rather than punitive. Nudges are non-blocking toasts you can snooze; Boundary warns instead of hard blocking; Buddy check-ins carry no leaderboard; Focus/Calm states ease transitions rather than snapping. Hi-fi designs for this Blueprint Design stage should preserve that tone.

Special case: Adapt to the Workspace is the one pillar where the design system itself needs to flex — it requires three distinct visual density states (Standard / Focus-Simplified / Calm-Recovery) rather than one fixed layout.

12. Assumptions, Constraints and Risks

Assumptions

- All demo data is synthetic and pre-generated (20 employees, 4 teams, 30 days) — no live production data is involved anywhere in this build.
- The team will continue splitting the six pillars by branch within the single shared repository, rather than separate repos per pillar.
- The Unified Employee Profile can begin once 2-3 pillars are built — not yet reached, since all six pillars are still at the Designed stage.

Constraints

- No real Slack/Calendar/Email API integrations are in scope for Boundary or Focus — both must remain fully simulated against synthetic data.
- Design tokens (hex values, inter font) are fixed and must match exactly between the hi-fi prototype and the production build.
- main is protected; every pillar's schema change must live in its own numbered migration file, and branches must pull main before starting new work.

Risks

- Cross-pillar dependency risk: Focus Mode's baseline sensitivity depends on Burnout's risk_scores.band; any change to Burnout's scoring logic after Focus Mode is built will require re-validating Focus Mode too.
- Migration-collision risk: seven migration files across up to six parallel branches is manageable but not collision-proof if two pillars land out of sequence.
- Privacy-enforcement risk: Mood, Buddy Check-ins, and Focus Mode all promise privacy "at the query level," a stronger and easier-to-violate-by-accident guarantee than UI-level hiding — each needs an explicit test, not just a design review.
- Schedule risk: all six pillars are still at the Designed stage — none have been built yet — with this Blueprint Design stage due Aug 10, 2026; build time for the entire system falls into whatever stage follows, so build velocity after this deadline is critical.
- Ownership risk: only two pillars currently have a named owner (Burnout — Charles Erick S. Ramos; Quick Nudge Tool — Gwychi); the remaining four have no assigned owner yet.

13. Timeline and Milestones

Phase	Deliverable	Target / Status
Phase 1 — Blueprint Design	This PRD + hi-fi prototypes for all six pillars, using shared design tokens	Due Aug 10, 2026
Phase 2 — Build	Shared core schema + Burnout Risk Analytics + Quick Nudge Tool	TBD — follows Stage 2
Phase 3 — Build	Track the Mood + Boundary Boundary	TBD
Phase 4 — Build	Give me a Coffee + Adapt to the Workspace (Focus Mode needs Burnout's band output stable)	TBD
Phase 5 — Unified Employee Profile	Joins all built pillars into /employees/[id]	Unlockable once Phase 3 completes

Exact calendar dates beyond the Aug 10, 2026 Stage 2 deadline are not yet set — see Section 14, Open Questions.

14. Open Questions

- What are the exact calendar dates for the Stage 3 build/submission of each pillar, beyond the Aug 10, 2026 Stage 2 design deadline?
- What is the hosting/deployment target for the demo — local dev only, or a deployed URL — for judging?
- Will the Centralized Employee Profile be attempted within this hackathon's timeline, given it depends on 2-3 pillars being built first and none are built yet?
- Are hi-fi prototypes (e.g., Figma or code) required alongside this PRD for the Stage 2 "BlueprintDesign" checklist item, or does this document alone satisfy it?

For the hi-fi link visit: <https://axion-hr-hackathon.vercel.app/>